

### ■ A.3.2 Laplace Transforms

Laplace transforms are a common type of integral transforms. They are particularly important in “operational methods” for solving partial differential equations.

You can define Laplace transforms in *Mathematica* by giving almost directly the formulae for Laplace transforms that you would find in a standard mathematical handbook. Each formula is implemented as a transformation rule for a pattern involving the object `Laplace`.

```
Laplace::usage = "Laplace[expr,t,s] gives the Laplace transform of expr."
```

```
(* constants *)
```

```
Laplace[c_,t_,s_] := c/s /; FreeQ[c,t]
```

```
(* linearity *)
```

```
Laplace[a+b_,t_,s_] := Laplace[a,t,s] + Laplace[b,t,s]
```

```
(* pick off constants *)
```

```
Laplace[c_ a_,t_,s_] := c Laplace[a,t,s] /; FreeQ[c,t]
```

```
(* powers *)
```

```
Laplace[t_^n_,t_,s_] := n!/s^(n+1) /;
                        (FreeQ[n,t] && n > 0)
```

```
(* products involving powers *)
```

```
Laplace[a_ t_^n_,t_,s_] :=
  (-1)^n D[Laplace[a,t,s], {s, n}] /; (FreeQ[n,t] && n > 0)
```

```
(* negative powers *)
```

```
Laplace[a_/t_,t_,s_] :=
  Block[ { v = Unique["s"] },
    Integrate[Laplace[a,t,v],{v,s,Infinity}]
  ]
```

```
(* exponentials *)
```

```
Laplace[a_ Exp[b_ + c_ t_],t_,s_] :=
  Laplace[a Exp[b],t,s-c] /;
  FreeQ[{b, c},t]
```

Laplace transforms.

Read in the file of Laplace transforms.

```
In[1]:= <<Laplace.m
```

Here is the Laplace transform of  $t^2 e^{-2t}$ .

```
In[2]:= Laplace[t^2 Exp[-2t], t, s]
```

$$\text{Out}[2]= \frac{2}{(2 + s)^3}$$

If you want *Mathematica* to find Laplace transforms of Bessel functions, you will have to give some more rules.

```
In[3]:= Laplace[BesselJ[0, t], t, s]
```

```
Out[3]= Laplace[BesselJ[0, t], t, s]
```